Macroeconomics I — MPE Insper

Practice list: The New Keynesian AS–AD Model (Lista 7)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.
Notation (Benigno 2015). Lower-case letters are logs. AD: $y = \bar{y}_n + (g - \bar{g}) - \sigma[\, i - (\bar{p} - p) - \rho\,]$ (eq. 21, no consumption taxes). AS: $p - p^e = \kappa(y - y_n)$ (eqs. 17, 20). Natural output $y_n = \frac{1+\eta}{\sigma^{-1}+\eta} a + \frac{\sigma^{-1}}{\sigma^{-1}+\eta} g - \frac{1}{\sigma^{-1}+\eta}\mu$ (eq. 15); efficient output y_e is the same without the μ term (eq. 19). Bars denote the long run.

Block 1 — Foundations: AD, AS, natural and efficient output *(Benigno §§1–5 • 25 points)*

Question 1. (10 points) Objective items.

a) (3 points) If the share of firms whose prices are fixed in advance rises, then:

A. κ rises, the AS gets steeper, and a cut in i moves output by more.

B. κ falls, the AS gets flatter, and a cut in i moves output by more.

C. κ is unchanged, the AS shifts up, and a cut in i moves prices by more.

D. κ falls, the AS gets steeper, and a cut in i moves prices by more.

b) (3 points) With $\sigma = \eta = 1$, a temporary rise of 0.04 in the (log) desired markup μ changes $(y_n,\, y_e)$ by:

A. $(-0.04,\, -0.04)$

B. $(0,\, -0.02)$

C. $(-0.02,\, 0)$

D. $(-0.02,\, -0.02)$

c) (2 points, T or F) In Benigno's model the AD slopes down because a higher price level raises money demand and therefore the nominal interest rate.

d) (2 points, T or F) After a permanent productivity increase, with i unchanged, the economy ends up at $p = p^e$ and $y = y_n$: both curves shift and no policy action is needed.

Question 2. (15 points) Discursive. Copom's model. Take $\sigma = \eta = 1$, $\kappa = 0.5$, $\rho = 0.02$, $p^e = \bar{p} = 0$, $\bar{y}_n = 0$, $g = \bar{g}$, and initially $a = \mu = 0$, so $i = \rho$ gives $y = p = 0$.

a) (5 points) Solve AD and AS for equilibrium y and p as functions of i, $\bar{p}$, p^e, y_n and $\bar{y}_n$. Which variables move AD and which move AS?

Solution:

b) (5 points) A temporary productivity gain: $a = 0.03$, $\bar{a} = 0$. Compute y_n, then y, p and the gap with $i = \rho$. Which i restores $p = p^e$ and $y = y_n$? Show that it violates the zero lower bound, compute the outcome at $i = 0$, and find the long-run price level $\bar{p}$ that the central bank must credibly promise to reach the target with $i = 0$.

Solution:

c) (5 points) A temporary markup shock (oil, a tax on firms): $\mu = 0.04$, $a = 0$. Compute y_n, y_e and the outcome with $i = \rho$. Find i under strict price stability. The bank minimizes $\mathcal{L} = (p - p^e)^2 + \lambda(y - y_e)^2$ subject to the AS, with $\lambda = 0.25$: derive the optimal y, p and i. Draw the diagram.

Solution:

Block 2 — Policy analysis

(Benigno §§6–12 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) The Focus survey suddenly expects higher *future* productivity; current productivity is unchanged (Benigno §6.3). The policy that keeps $p = p^e$ and $y = y_n$ is to:

A. raise i, since AD shifts right while AS stays put.

B. cut i, since AS shifts right while AD stays put.

C. keep i, since y_n today is unchanged by news.

D. promise a higher $\bar{p}$, since AD shifts left today.

b) (3 points) With $\sigma = \eta = 1$, $\kappa = 0.5$ and i fixed, a temporary rise of 0.01 in g raises output by:

A. 0.0100

B. 0.0050

C. 0.0150

D. 0.0083

c) (2 points, T or F) At the zero lower bound, monetary policy has no instrument left to raise current output.

d) (2 points, T or F) Productivity and government-spending shocks move y_n and y_e by the same amount, so they create no trade-off between stabilizing prices and the efficient gap.

Question 4. (15 points) Discursive.

a) (5 points) Derive y_n and y_e from first-order conditions in log form. Use the labor supply condition $\eta l + \tilde{\sigma}^{-1} c = w - p$, the price rule of flexible firms $p = w - a + \mu$, the technology $y = a + l$ and the resource constraint $y = s_c c + g$, with $\sigma = \tilde{\sigma} s_c$. For y_e, replace the price rule by the planner's condition "marginal rate of substitution = marginal product of labor". Explain why a markup shock moves y_n but not y_e.

Solution:

b) (5 points) The Copom cuts the Selic by one percentage point. Compare the short- and long-run effects predicted by (i) the New Classical model, (ii) the original Keynesian (IS–LM) model and (iii) Benigno's New Keynesian model, along three dimensions: price adjustment and expectations; the policy instrument and how AD is built; and whether the model can rank outcomes by welfare.

Solution:

c) (5 points) A deleveraging shock (Eggertsson–Krugman): the debt limit of borrowers falls sharply. Explain why the economy can fall into a liquidity trap, why the fiscal multiplier becomes large, and why more price flexibility can make the recession worse (paradox of flexibility). Under what condition on long-run prices does the paradox disappear?

Solution: